

TEGAN JEGEDE

AI Research Scientist & Autonomous Agents Engineer.

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EDUCATION

NILE UNIVERSITY ABUJA, NIGERIA

Masters of Science - Computer Science

Oct 2024 – Ongoing

- **Research Thesis:** *"Mitigating Hallucination in Agentic Systems via Hybrid Reward Architectures."*
- **Focus Areas:** Autonomous Agents, Reward Modeling, AI Safety, and Multi-turn LLM Interactions.

UNIVERSITY OF ABUJA, ABUJA, NIGERIA

Bachelor Of Engineering- Electrical & Electronics Engineering

Nov 2015 – Feb 2022

RESEARCH AND ENGINEERING EXPERIENCE

NILE UNIVERSITY ABUJA

AI Researcher (MSc Thesis)

October 2024 – Present

- **Reward Modeling & Safety:** Designing hybrid reward architectures to penalize non-factual reasoning steps in LLM powered autonomous agents, directly addressing hallucination in open-ended tasks.
- **Evaluation Protocols:** Developing evaluation metrics to capture the success of complex multi turn interactions, assessing behavioral trajectories and implicit user state inference.
- **Empirical Validation:** Conducting rigorous ablation studies to benchmark agent performance against baseline models, analyzing failure modes to iteratively refine safety guardrails.

Research intern(NLP & LLM Prototyping).

Oct 2024 – Present

- **Semantic Pipelines:** Engineered production-grade NLP pipelines leveraging LangChain, spaCy, and NLTK to extract and evaluate candidate information from unstructured data.
- **Model Evaluation:** Gained hands-on experience evaluating frontier Large Language Models (GPT-4, LLaMA 2, Mistral) for real-world information extraction use-cases.
- **Data Analysis & ML:** Conducted Exploratory Data Analysis (EDA) and trained ensemble models/Decision Tree Regressors using Scikit-Learn, achieving 70% accuracy on complex predictive tasks.

TECHNOLOGY EDUCATION SPECIALIST(Digital Skills Trainee)

E-suffers paradise

Feb 2024- present

- Pioneered and executed training strategies across Kogi and Kwara states as part of a Federal Government funded initiative to equip 200,000 Nigerians in rural areas with essential digital skills.
- Delivered hands on and online training focusing on core competencies including Microsoft Office, Google Workspace, Data Analysis, and Cloud/AI fundamentals to diverse groups, including high school teachers and students.
- Analyzed and addressed the digital skill gap in the Middle Belt region, informing program refinement and directly supporting the project's goal of training over 150,000 high school students and 750 teachers.

PRODUCT MANAGER (Remote)

ScaleInAfrica

Aug 2022 – Jan 2024

- Drove Scrum and Agile best practices across a remote 10 person engineering team, resulting in clearer sprint goals and a reduction in scope creep.
- Facilitated all key Scrum ceremonies (Daily Stand ups, Sprint Planning, Retrospectives), refocusing the Daily Stand ups to preemptively flag blockers and improve cross functional transparency.
- Established a standardized communication flow using Jira and Slack to maintain alignment across stakeholders and engineering, minimizing context switching for the development team.

PROJECTS

AI Agent for Social Media Management

Python, Agno, Streamlit, Groq

Developed an autonomous AI agent for social media managers, leveraging Agno for robust agentic workflows and tool orchestration. Built an intuitive, real time user interface with Streamlit, powered by Groq for lightning fast LLM inference. Automated tasks such as content scheduling, engagement analysis, and personalized response generation to optimize social media strategies.

Retrieval-Augmented Generation (RAG) System

Python, LangChain, Streamlit and groq

Engineered a RAG system leveraging LangChain to integrate external knowledge bases (MSc lecture notes) with Large Language Models (LLMs), enhancing factual accuracy and reducing hallucinations for studying and summarization. Developed an interactive web application using Streamlit for seamless user interaction and real time querying of custom data.

Final Year – Comparison of Load demand Forecasting Between Machine Learning Model and Linear Regression Models in Abuja, Nigeria.

SKILLS

- **AI/ML & Frameworks:** PyTorch, OpenAI Gymnasium (RL), LangChain, LangGraph, Agno, LlamaIndex, Scikit-Learn, Hugging Face, Keras.
- **Core Languages & Tools:** Python, SQL, Dart, MATLAB, Jupyter, Git, Firebase.
- **Domain Expertise:** Autonomous Agents, Reward Modeling concepts, Multi-Agent Systems, NLP, Prompt Engineering, End-to-End Prototyping, Agile/Scrum.

CERTIFICATIONS

- NIIT Abuja -Python programming language certification.
- Terraskills institute Abuja - Data analysis certification.
- Skylight institute Abuja - Project management certification.